

IOT BASED SMART WATER PUMP SYSTEM

Abstract:

This project aims to provide an efficient solution for controlling a water pump remotely using IoT technology. ESP32 and Blynk app are used to switch ON/OFF the pump using a smartphone. This eliminates the need for the user to always be physically present near the pump. To improve the system performance further, a soil moisture sensor is additionally integrated, which can automatically operate the pump whenever the soil becomes dry. Thus, the system provides both manual IoT control and automatic smart irrigation capability, offering better water management and user convenience.

Introduction:

In modern agriculture and gardening, maintaining correct soil moisture is essential for plant growth. Traditional pumping methods require regular observation and manual operation which consumes time and may lead to water wastage. With advancements in IoT, remote monitoring and controlling of devices have become easier. The ESP32 microcontroller includes in-built Wi-Fi capability, making it suitable for IoT automation.

In this project, Blynk platform is used to control the pump wirelessly. In addition, a soil moisture sensor has been introduced as a value-added feature to automate the watering system when required. This dual functionality makes the project more smart, efficient, and useful for different applications.

Objectives:

- To remotely control a pump using Blynk IoT platform.
- To reduce the user's physical effort and ensure convenience in watering.
- To improve irrigation efficiency by adding moisture-based automatic control.
- To reduce unnecessary water wastage and promote smart farming technology.

Components Required:

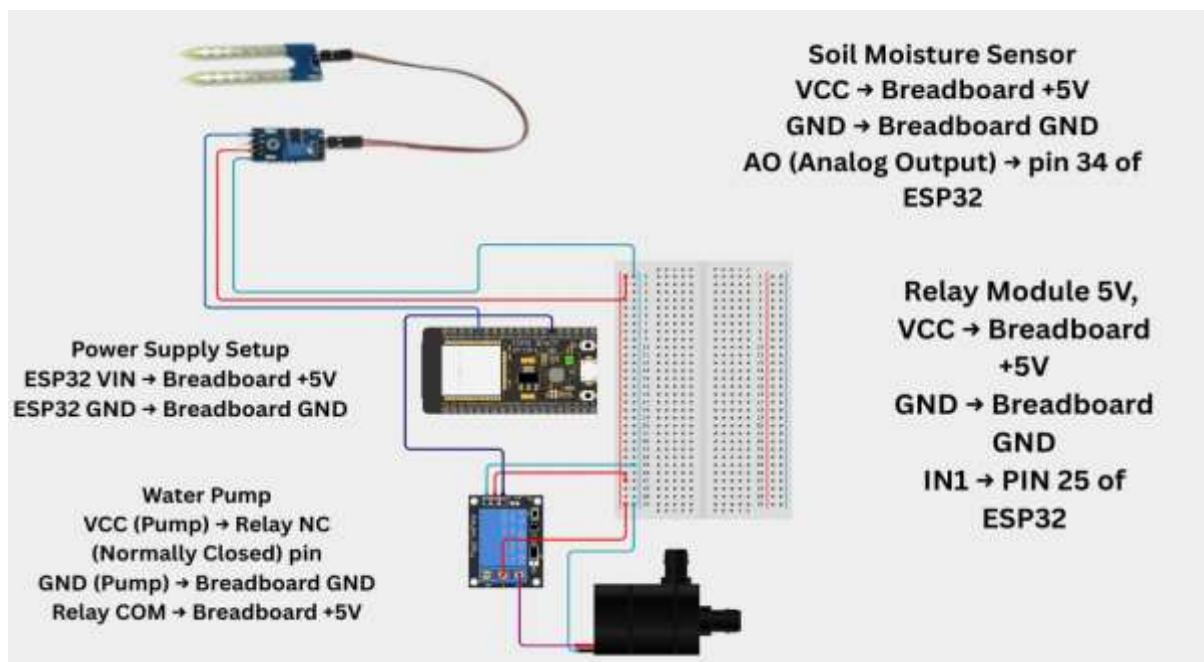
- ESP32 Development Board.
- Relay Module (5V, Single Channel).
- 5V Water Pump.
- Soil Moisture Sensor (Optional Enhancement).
- Blynk Mobile App / Wi-Fi Router.
- Breadboard & Connecting Wires.
- External Power Supply for Pump.

Circuit Description:

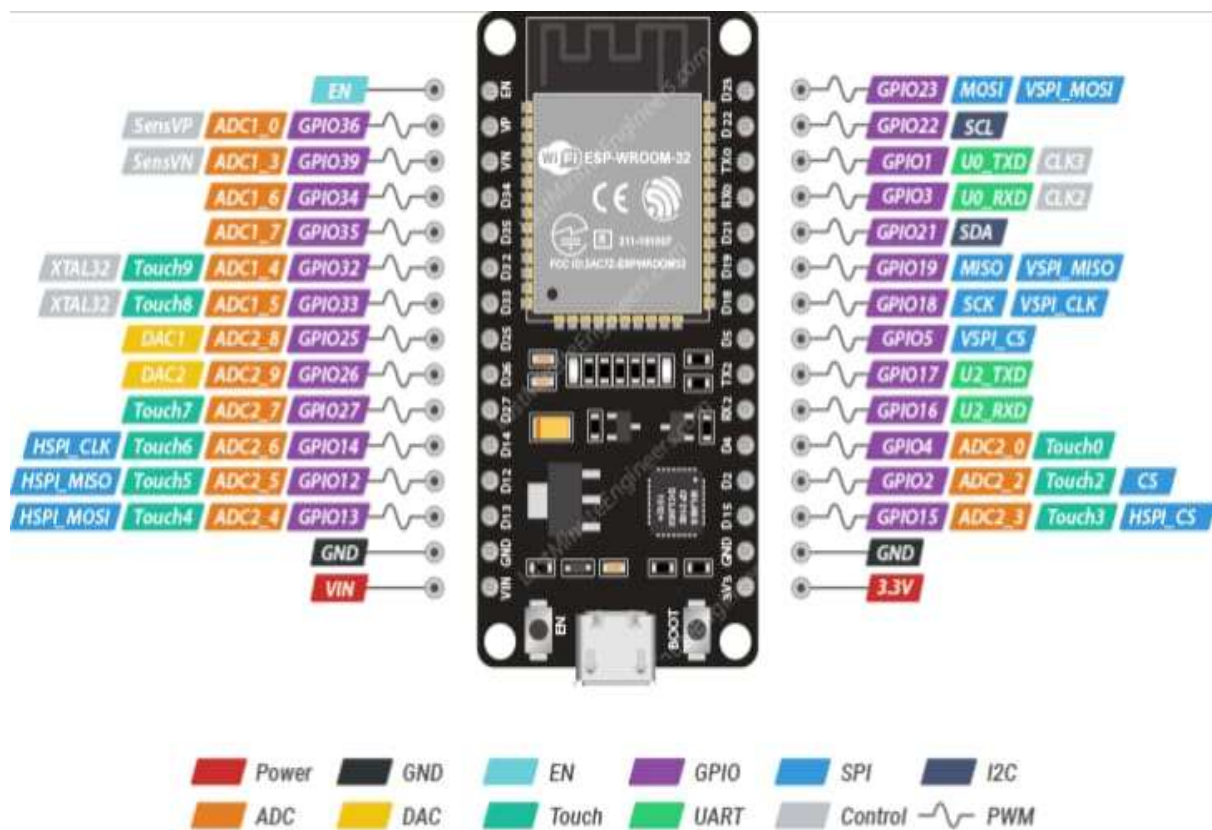
The relay module acts as a switching interface between the ESP32 and the pump, providing electrical isolation to ensure component safety. The ESP32 GPIO pin drives the relay input and controls the pump status. The moisture sensor is connected to an analog pin for continuous data reading.

Power supply lines are distributed through a breadboard for easy connections. The ESP32 is powered with 5V through the VIN pin and grounded properly to avoid electrical noise. Proper wiring ensures smooth and safe operation of the pump without interference.

Circuit Diagram and Connection:



ESP 32 pinout:



ESP 32 Detailed Description:

Introduction to ESP32

The ESP32 is a highly advanced microcontroller developed by Espressif Systems, specially designed for modern IoT (Internet of Things) applications. In our IoT Based Smart Pump Control and Monitoring System, the ESP32 acts as the main controller that connects to the internet using its in-built Wi-Fi. The ESP32 receives commands from the Blynk mobile application over the cloud and accordingly controls the water pump through a relay module.

Because of its wireless capability and low-cost architecture, it removes the need for manual switches near pumps, allowing safe and remote operation.

Additionally, it can read input from a soil moisture sensor and automatically turn ON/OFF the pump without human intervention. Due to these advantages, ESP32 is widely used in smart irrigation, home automation, and wireless sensor networks.

Features of ESP32:

- The ESP32 includes several important features that make it suitable for automation applications like our smart pump:

- Built-in Wi-Fi (802.11 b/g/n)
- Helps to connect ESP32 with the Blynk cloud for remote pump control.
- Dual-Core 32-bit Processor.
- Handles both wireless operations and real-time control tasks simultaneously.
- Built-in Bluetooth & BLE.
- Used for local wireless monitoring without internet.
- Multiple GPIO Pins.
- Used to connect relay, sensors and other control devices.
- Analog to Digital Converter (ADC).
- Reads soil moisture sensor output to enable automatic irrigation.
- Low Power Modes.
- Helps in saving power when used in remote agricultural locations.
- High Processor Speed (up to 240 MHz).
- Ensures fast response between command and pump action.
- PWM, I2C, UART, SPI Protocols
- Allow future expansion with more sensors such as flow, temperature etc.

ESP32 Architecture:

The ESP32 architecture consists of:

- CPU
- Dual core Xtensa processor that divides workload between Wi-Fi tasks and sensor/pump control tasks.
- Example: One core handles Blynk commands while the other manages the relay and moisture sensor.
- Wi-Fi & Bluetooth Radio.
- Provides wireless cloud connectivity ideal for smart irrigation systems.
- Internal Memory (RAM & Flash)
- Stores program code and Blynk libraries supporting IoT operation.
- Power Management Unit
- Allows ESP32 to work on solar or battery in farms with power-saving modes.
- Security Engine
- Provides secure encrypted communication with Blynk cloud to avoid unauthorized pump control.
- Analog Interface (ADC)
- Reads wetness level of soil from sensor and decides pump operation.
- Digital Input/Output Module (GPIO)

- Controls the relay to turn ON/OFF the pump and can detect switch feedback if needed.
- This architecture ensures that the ESP32 can handle continuous monitoring and long-distance operation safely and efficiently.

ESP32 Pin Description:

- The ESP32 has several pins, but only essential pins are used for our Smart Pump System:
- Pins Used in Project:
- 3.3V Pin Power supply for moisture sensor.
- GND Pin Common ground reference for all components.
- GPIO Pin for Relay Control (Example: GPIO 5/23 etc.) Sends digital HIGH or LOW signal to operate pump relay.
- ADC Pin (Example: GPIO 34, 35, 32 etc.) Reads analog value from soil moisture probe.

Additional Useful Pins

- UART Pins (TX/RX) – For future debugging and serial monitoring.
- I2C Pins (SDA, SCL) – Can be used to add LCD or digital sensors like DHT11.
- Boot / EN Pin – Used only during programming or reset.

Power Requirements of ESP32:

ESP32 operates at:

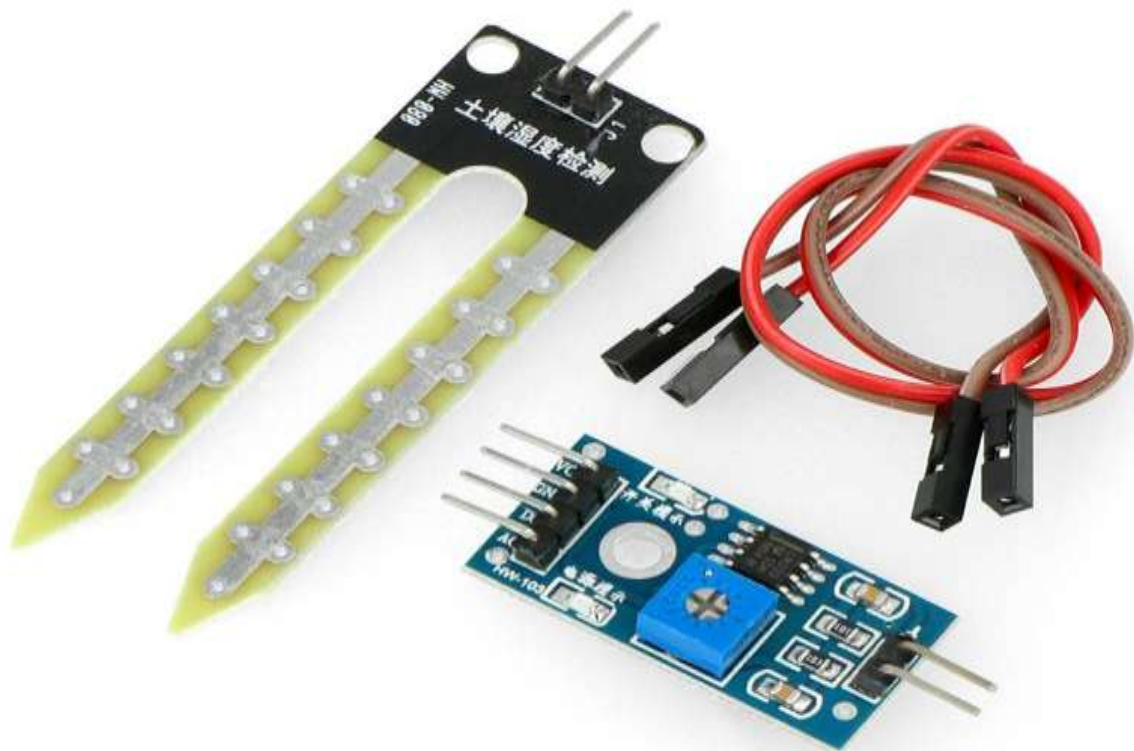
- 3.3V logic level (essential for safe sensor connections)
- VIN Pin supports 5V USB power.
- Normal operating current: 80–240 mA during Wi-Fi transmission.
- Thus, it can be easily powered from a phone charger or 5V adapter in farm sheds.

Advantages of Using ESP32

- Reduces human effort & water wastage.
- Ensures timely irrigation with crop health improvement.
- Enhances safety since farmer need not go near pump.
- Very low cost compared to commercial IoT systems.
- Future-upgradable system with sensors and alarms.

Details Of Soil Moisture Sensor:

Soil Moisture sensor:



Introduction:

Soil moisture is one of the most essential physical parameters necessary for the growth of plants. It directly determines the water availability in soil. In modern agriculture, it becomes very important to measure the amount of moisture present in soil accurately so that automatic irrigation systems can be used to save water and increase crop productivity. In traditional farming, farmers manually check soil conditions which leads to delay and wastage of water. Therefore, soil moisture monitoring using electronic sensors plays a major role in Smart Farming and IoT-based Agriculture.

Working Principle of Soil Moisture Sensor:

The soil moisture sensor generally works based on the property of soil to conduct electricity. Wet soil has high water content, so it allows more electric current to flow. On the other hand, dry soil restricts current flow due to less

water content. By measuring how easily current flows through the soil, we can determine how much moisture is present.

- Wet Soil → High Conductivity → Low Resistance
- Dry Soil → Low Conductivity → High Resistance

This change in resistance is converted into a measurable voltage which is used as the output signal.

Some sensors use resistive technology, while others use capacitive measurement, where soil's dielectric constant changes due to moisture variation.

Types of Soil Moisture Sensors:

There are mainly two commonly used types of soil moisture sensors in electronic projects:

1. Resistive Soil Moisture Sensor

- Uses 2 metal probes inserted into soil.
- Measures resistance/conductivity.
- Low cost and widely used in hobby projects.
- Requires frequent maintenance due to probe corrosion.
- Output can be analog or digital.

2. Capacitive Soil Moisture Sensor

- Uses change in dielectric permittivity of soil.
- No direct contact of wires with soil → No corrosion.
- More stable and long-lasting.
- Only analog output, but accurate and reliable.

Pin Configuration :

- Most soil moisture sensors provide 3 or 4 pins:
- VCC → Power supply (3.3 to 5V).
- GND → Ground.
- AO → Analog Output (0–5V).
- DO → Digital Output (optional in some types)

A basic soil moisture sensing system includes:

1. Soil moisture sensor probe
2. Signal conditioning circuit (Amplifier/Comparator)
3. Microcontroller input (ADC or Digital pin)
4. Display or Pump control mechanism

Advantages Of Soil Moisture Sensor:

- Helps in saving water efficiently.
- Automates irrigation system → Reduces human effort.
- Improves plant growth and crop yield.
- Easy to interface with microcontrollers.
- Low cost and compact.

Applications Of Soil Moisture Sensor:

- Smart garden and home automation.
- Agricultural irrigation systems.
- Greenhouse monitoring.
- Soil testing in nurseries.
- Environmental monitoring systems.
- Smart IoT-based farming solutions.

Electrical Characteristics (General Specs)

Feature	Typical Value
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Supply Voltage	-3.3V – 5V
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Average Current	- 10 to 20 mA
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Output Voltage	- 0 – 5V
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Operating	
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Temperature	- -10°C to +60°C
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Probe Material - Metal (Nickel plated) / PCB type

Working in IoT Systems:

This sensor is commonly used in:

- ESP32 / NodeMCU with cloud monitoring.
- Blynk or ThinkSpeak IoT apps.
- Automatic pump control based on soil moisture level

Limitations:

- Resistive type sensors face probe corrosion when used continuously in wet soil.
- Not highly accurate for industrial agriculture without calibration.
- Soil type changes output, so calibration is needed.
- Capacitive sensors reduce these problems but cost slightly more.

Detailed Description of 5V relay:



Introduction to Relay Module:

A relay is an electromagnetic switch used to control a high-power electrical device using a low-power signal. Microcontrollers like ESP32 operate at 3.3V logic, which is not enough to directly drive motors or pumps.

So, a relay module acts as a safety and isolation interface between low-voltage control electronics and high-voltage load.

In our IoT Based Smart Pump System, the relay module is used to turn ON/OFF the water pump based on instructions received from the ESP32/Blynk or based on soil moisture level. Thus, it enables the automation of pump operation safely and efficiently.

Relay ensures complete electrical isolation using optocouplers or electromagnetic switching, protecting ESP32 from back EMF and voltage spikes caused by the pump.

Working Principle of Relay Module:

A relay contains:

- An electromagnetic coil.
- A mechanical armature.
- Switching contacts (COM, NO, NC pins)

When ESP32 gives a HIGH logic to the relay input:

1. Current flows through the relay coil
2. Magnetic field is generated
3. Armature is pulled and switching happens
4. Pump turns ON/OFF according to connection

Relay Contact Explanation:

Pin	Full Form
COM	- Common Terminal
NO	- Normally Open
NC	- Normally Closed

Why Relay is Used in Smart Pump IoT System?

- Pump requires higher current than microcontroller can provide.
- Relay safely isolates ESP32 from 5V pump load.
- Ensures stable switching even for inductive loads (pumps).
- Allows remote ON/OFF control through mobile app.
- Supports automatic switching using moisture sensor data.
- Prevents reverse current damage using protection circuitry.
- Thus relay makes the project safe, durable & industrial grade.

Interfacing Relay with ESP32:

In the proposed system:

- Relay IN1 pin → GPIO 25 of ESP32
- Relay VCC → 5V supply.
- Relay GND → ESP32 Ground.
- Pump connected via COM & NO terminals

Signal Characteristics:

- Relay input responds to 3.3V logic from ESP32.
- Digital HIGH = Pump ON.
- Digital LOW = Pump OFF

Advantages of Relay-Based Pump Automation:

- Electrically isolates sensitive controller from pump.
- Safe switching of high-current loads.
- Very low power consumption for control.
- Easy to interface and program.
- Improves irrigation efficiency.
- Supports both manual & automatic control logic.
- Reliable for long-term field usage

Principle Of Operation:

The main principle of this project is IoT based switching. The ESP32 connects to Wi-Fi and communicates with Blynk cloud servers. When the user presses a switch button in the Blynk app, the ESP32 receives the command and controls the relay, which in turn switches the water pump ON or OFF.

Upgraded Principle: A soil moisture sensor is additionally used to monitor soil dryness. The microcontroller reads the sensor value and, if soil moisture is too low, automatically turns ON the pump. Once the soil becomes sufficiently wet, the pump is turned OFF again. This improves automation and reduces manual supervision.

Working Process:

1. Using ON/OFF control from Blynk:

- User can ON/OFF the water pump from anywhere via smartphone.
- The relay is directly controlled based on the signal from the Blynk dashboard.
- The moisture sensor does not affect water pump operation in this mode.
- Best suited when user wants full personalized control over watering.

2. Using Soil Moisture Sensor:

- The soil moisture sensor continuously reads soil condition.
- If moisture level is less than 30, the water pump turns ON automatically.
- If moisture level becomes greater than 30, the water pump turns OFF automatically.
- No user action needed, ensures efficient irrigation management.

Software Implementation:

- Arduino IDE is used to upload the program into ESP32.
- ESP32 Wi-Fi library enables internet connectivity.
- Blynk library manages cloud communication and virtual switch.
- A timer function is used for periodic sensor reading.
- Thus, both IoT and automation are executed through integrated software modules.

Code For Project:

```

#define BLYNK_TEMPLATE_ID "TMPL3dVVFTF51"
#define BLYNK_TEMPLATE_NAME "Smart Irrigation"
#define BLYNK_AUTH_TOKEN
"KWLCKxSJYEjSdo5RqDMU41Bg1IpAidxk"

#include <WiFi.h>

#include <BlynkSimpleEsp32.h>

char ssid[] = "VIKNESH RAJ";
char pass[] = "33333333";

#define SOIL_SENSOR_PIN 34
#define RELAY_PIN 25

int moistureLevel = 0;
int timerSeconds = 0;
unsigned long pumpOffTime = 0;
bool manualRunning = false;
BlynkTimer timer;
BLYNK_WRITE(V2)
{
    timerSeconds = param.asInt();
}
BLYNK_WRITE(V0)
{
    int btn = param.asInt();
    if (btn == 1)
    {

        digitalWrite(RELAY_PIN, HIGH);
        manualRunning = true;
        if (timerSeconds > 0)

```

```

{
  pumpOffTime = millis() + (timerSeconds * 1000);
}
else
{
  pumpOffTime = 0;
}
}
else
{
  digitalWrite(RELAY_PIN, LOW);
  manualRunning = false;
  pumpOffTime = 0;
}
}
void checkSoil()
{
  moistureLevel = analogRead(SOIL_SENSOR_PIN);
  int percentage = map(moistureLevel, 0, 4095, 100, 0);
  Blynk.virtualWrite(V1, percentage);
  if (!manualRunning)
  {
    if (percentage < 30) {
      digitalWrite(RELAY_PIN, HIGH);
    } else {
      digitalWrite(RELAY_PIN, LOW);
    }
  }
}

```

```

    }
}

void checkPumpTimer()
{
    if (manualRunning && pumpOffTime > 0)
    {
        if (millis() >= pumpOffTime) {
            digitalWrite(RELAY_PIN, LOW);
            manualRunning = false;
            pumpOffTime = 0;
            Blynk.virtualWrite(V0, 0);
        }
    }
}

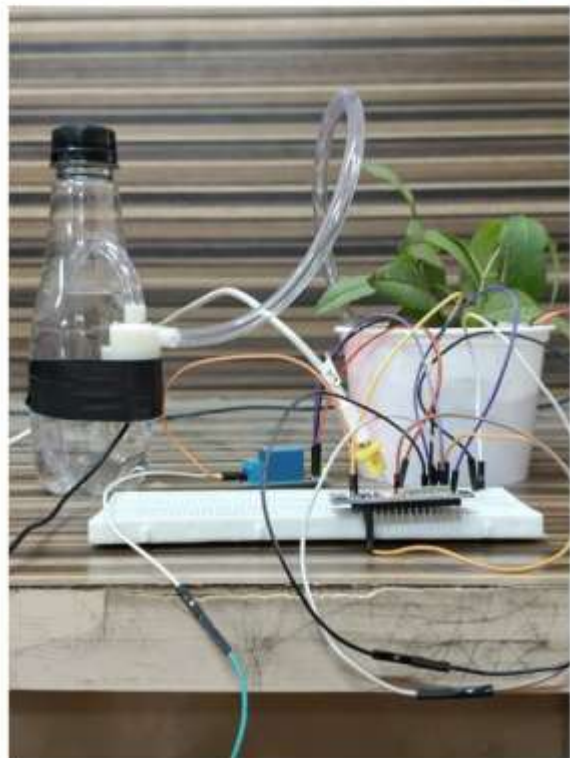
void setup()
{
    Serial.begin(115200);
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW);
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
    timer.setInterval(3000L, checkSoil);
    timer.setInterval(500L, checkPumpTimer);
}

void loop()
{
    Blynk.run();
    timer.run();
}

```

}

Hardware Setup:



Conclusion:

The IoT based smart pump system successfully demonstrates remote water pump control using ESP32 and Blynk. The additional soil moisture sensing feature enhances the project by allowing automatic irrigation. Since both functions operate independently, the system can be used in multiple situations depending on user needs. This project highlights the importance of IoT in making farming operations easier and more intelligent.

Future Enhancements:

- Cloud data storage and water consumption analytics.
- Solar power integration for rural areas.
- Multi-zone irrigation using multiple sensors.
- Real-time notifications to user's mobile.
- Weather forecast-based watering logic.